

13-Bus Transmission N-1 Contingency Analysis Using PowerWorld and Tableau

Power Flow Modeling, Contingency Testing, Corrective Action Planning, and Dashboard-Based Visualization for a Simplified Ontario Inspired Transmission Network

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Abstract

This project analyzes the reliability of a simplified 13-bus transmission network under N-1 contingency conditions. A base case power flow model was created in PowerWorld Simulator and tested by removing one transmission element at a time. For each outage, voltage violations, thermal overloads, maximum branch loading, minimum bus voltage, and recommended corrective actions were recorded.

Of the nine contingency cases tested, five were classified as severe and four as moderate. The most critical case, the loss of the Bruce Generation to GTA West connection, produced a minimum system voltage of 0.645 pu and a maximum branch loading of 305.5 %, indicating a heavy dependence on that transfer path with insufficient alternate support.

The results were summarized in structured CSV format and visualized using a Tableau dashboard to communicate system vulnerabilities and corrective planning decisions. The project demonstrates the workflow used in transmission planning studies, from model creation through violation identification, corrective action planning, and engineering communication.

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1 Introduction

Transmission systems must be able to maintain reliable operation after the loss of a single major component, such as a transmission line, transformer, or generator connection. This requirement is commonly evaluated using N-1 contingency analysis, which tests the system's ability to withstand the outage of any one element while continuing to serve load within acceptable voltage and thermal limits.

The purpose of this project is to model a simplified transmission network, perform N-1 outage testing, identify weak points in the system, and recommend corrective actions based on power flow results. The project combines three main components: PowerWorld Simulator for power flow modeling, structured CSV files for results tracking, and Tableau for dashboard based visualization. The goal is to demonstrate both technical power system analysis and practical engineering communication.

The project followed the workflow below:

Model setup → base case solution → N-1 outage testing → violation identification →
corrective action planning → dashboard visualization

2 Project Objective

The objective of this project is to evaluate how a simplified 13 bus transmission network behaves when individual branches are removed from service. The analysis focuses on:

- Identifying voltage violations after each N-1 outage.
- Identifying thermal overloads on transmission branches and transformers.
- Ranking contingencies by severity.
- Proposing corrective actions for overloaded or undervoltage conditions.
- Creating a Tableau dashboard to summarize the results.

3 System Description

The simulated system represents a simplified Ontario inspired transmission network. It includes generation sources, major load centers, transformer stations, an intertie bus, and GTA area transmission corridors. The model is not intended to represent the exact Ontario grid; instead, it is a simplified planning style network designed to test contingency analysis workflow and engineering interpretation.

3.1 Bus Summary

Table 1 summarizes the main buses in the system.

Table 1: Main system buses.

Bus	Role
Bruce Generation	Major generation source
Northern Hydro	Generation source
Intertie Bus	Generation / interconnection support
Peaker/Battery	Generation / fast response support
Western Ontario	Transfer bus
GTA West	Transfer bus
GTA Central	Major load center
Toronto Load	Major load center
Eastern Ontario	Transfer bus
Ottawa	Load center
Southwest Industrial	Load center
Transformer Station A	Transformer path
Transformer Station B	Transformer path

The model includes a mix of 500 kV and 230 kV buses, transmission lines, transformer paths, generation sources, and major load points. The system was designed so that removing key branches would create observable voltage and thermal stress, allowing N-1 contingency impacts to be studied.

4 PowerWorld Model Setup

The system was built in PowerWorld Simulator using bus, branch, transformer, generator, and load records. The base case was debugged until Newton Raphson power flow converged successfully.

4.1 Modeling Steps

The major modeling steps were:

1. Created the 13 bus transmission network.
2. Added generators at Bruce Generation, Northern Hydro, Intertie Bus, and Peaker/Battery.
3. Added major load centers at GTA Central, Toronto Load, Ottawa, Western Ontario, Southwest Industrial, and Eastern Ontario.
4. Added transmission lines and transformer connections between major buses.
5. Checked branch status, bus voltage levels, and branch limits.
6. Solved the base case using Newton Raphson power flow.
7. Performed N-1 outage cases by opening one branch at a time.

4.2 Base Case Power Flow

The base case was used as the reference operating condition before applying contingencies. After debugging the model, the system successfully solved using Newton Raphson power flow. The base case provided a stable starting point for comparing the effects of each N-1 outage. The purpose of the base case was not to optimize the system, but to establish a valid operating condition where all contingency cases could be tested consistently.

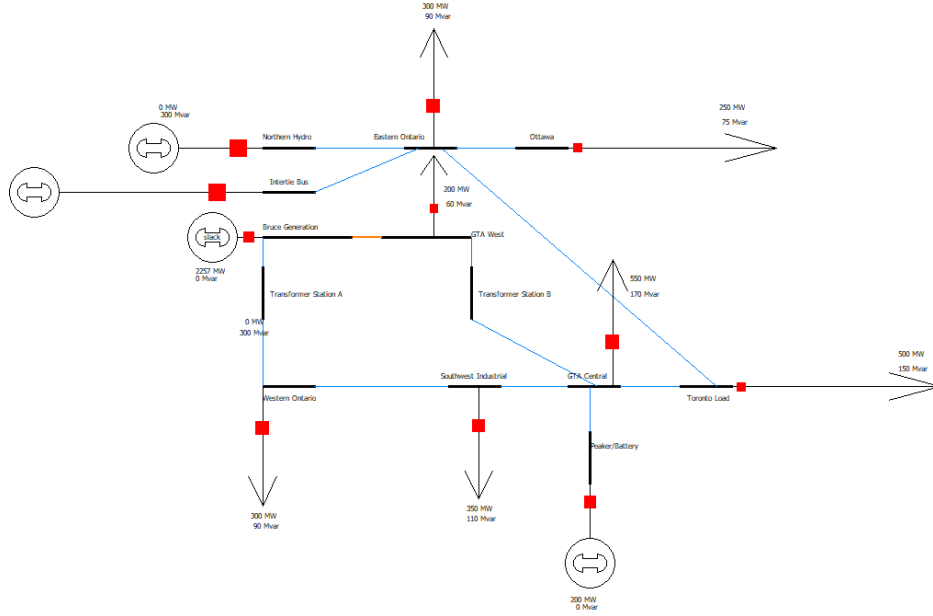


Figure 1: PowerWorld base case one-line diagram.

5 N-1 Contingency Methodology

For the N-1 analysis, each selected branch was opened individually while all other branches were returned to service. After each outage, the power flow was solved and the following values were recorded:

- Outaged element.
- Minimum bus voltage, in per unit.
- Bus with the lowest voltage.
- Maximum branch loading percentage.
- Branch with the highest loading.
- Primary violation.
- Severity classification.
- Recommended corrective action.
- Related screenshot file.

A contingency was classified as severe when it caused major undervoltage, severe thermal overloads, or both. A contingency was classified as moderate when the system remained solvable but still showed warning-level voltage or thermal stress.

6 N-1 Results

Table 2 summarizes the nine N-1 contingency cases.

Table 2: N-1 contingency summary results.

Case	Outaged Element	Min Voltage Bus	Max Loading Branch	Min V (pu)	Max Load (%)
N-1 Case 1	Bruce Generation to GTA West	GTA West	Transformer Station A to Western Ontario	0.645	305.5
N-1 Case 2	Bruce Generation to Transformer Station A	Western Ontario / Transformer Station A	Transformer Station B to GTA Central	0.851	186.9
N-1 Case 3	GTA West to Transformer Station B	Toronto Load	Transformer Station A to Western Ontario	0.833	275.5
N-1 Case 4	Transformer Station B to GTA Central	Toronto Load	Transformer Station A to Western Ontario	0.833	227.5
N-1 Case 5	GTA Central to Toronto Load	Toronto Load	Northern Hydro to Eastern Ontario	0.928	142.3
N-1 Case 6	Toronto Load to Eastern Ontario	Toronto Load	Transformer Station A to Western Ontario	0.964	111.0
N-1 Case 7	Northern Hydro to Eastern Ontario	Ottawa	GTA Central to Toronto Load	0.872	139.2
N-1 Case 8	Intertie Bus to Eastern Ontario	Ottawa	GTA Central to Toronto Load	0.941	165.5

Case	Outaged Element	Min Voltage Bus	Max Loading Branch	Min V (pu)	Max Load (%)
N-1 Case 9	Peaker/Battery to GTA Central	Ottawa	Transformer Station B to GTA Central	0.939	146.7

The most severe case was N-1 Case 1, where the outage of the Bruce Generation to GTA West connection caused the lowest system voltage of 0.645 pu and the highest observed branch loading of 305.5 %. This indicates that the system is highly dependent on this transfer path and does not have enough alternate support under that outage condition. Cases 3 and 4 also produced severe undervoltage and major overloads, showing that the GTA supply corridor and transformer station paths are critical parts of the network. Cases 5, 6, 7, and 9 were less severe but still showed voltage or loading concerns that would require operational attention.

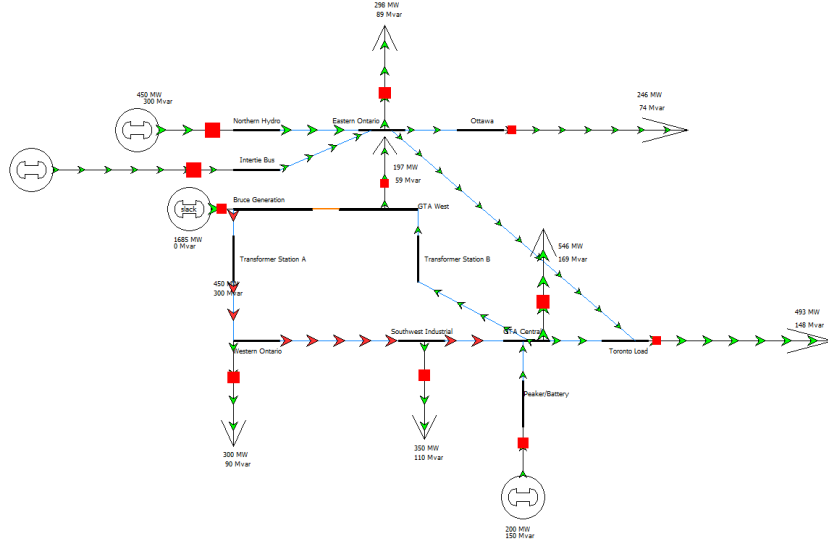


Figure 2: PowerWorld result for N-1 Case 1, Bruce Generation to GTA West outage.

7 Corrective Actions

Corrective actions were developed based on the observed violations in each contingency case. The recommended actions focused on reducing transfer stress, increasing local support, reinforcing overloaded paths, or adding alternate supply routes.

Table 3: Corrective action summary.

Case	Main Issue	Recommended Corrective Action
N-1 Case 1	Severe undervoltage and major overloads	Reduce Bruce transfer, dispatch local GTA/eastern support, and add or uprate alternate GTA supply path.
N-1 Case 2	Undervoltage and transformer overload	Uprate transformer path and increase local support near Western Ontario/GTA Central.
N-1 Case 3	Severe undervoltage and transformer overload	Add or uprate alternate GTA supply path and reduce transfer through Station A path.
N-1 Case 4	Undervoltage across GTA/eastern buses	Add or uprate alternate GTA Central supply path and reinforce Station A/Western corridor.
N-1 Case 5	Voltage warning at Toronto Load	Add or uprate alternate Toronto supply path and increase local voltage support.
N-1 Case 6	Moderate transformer overload	Redispatch generation to reduce western transfer and monitor transformer loading.
N-1 Case 7	Undervoltage in eastern buses	Increase eastern supply support and dispatch intertie or peaker support.
N-1 Case 8	Voltage warning and GTA overloads	Increase eastern support, reduce GTA/Toronto transfer stress, and uprate Toronto supply path.
N-1 Case 9	Loss of local GTA support	Dispatch replacement local support near GTA Central and reinforce Transformer Station B to GTA Central path.

The corrective actions show that the most important reinforcement areas are the GTA supply paths, transformer corridors, and eastern system support paths. These are the areas where

outages caused the largest voltage drops or branch overloads.

8 Tableau Dashboard

A Tableau dashboard was created to communicate the N-1 results visually. The dashboard includes maximum branch loading by contingency case, minimum bus voltage by contingency case, a severity breakdown of all N-1 cases, and a results table with corrective actions.

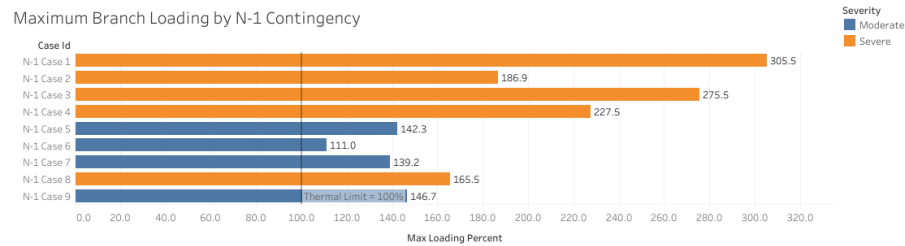
13-Bus Transmission N-1 Contingency Analysis Dashboard

PowerWorld simulation results showing thermal overloads, voltage violations, contingency severity, and recommended corrective actions.

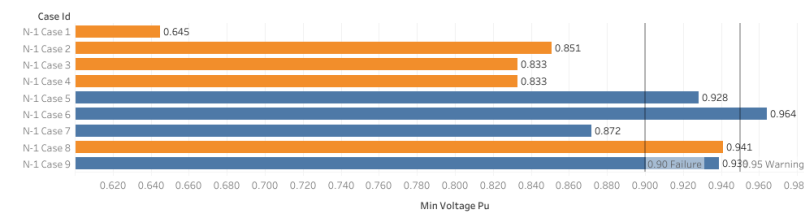
Key Findings:

- 5 of 9 N-1 contingencies were classified as severe.
- Case 1 caused the worst thermal overload at 305.5%.
- Case 1 caused the lowest voltage at 0.645 pu.
- The most stressed paths were the GTA supply and transformer corridors.

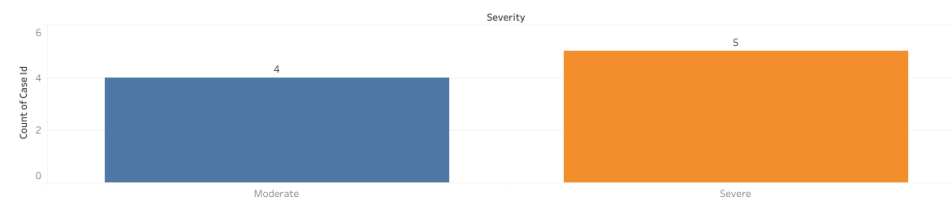
Maximum Branch Loading by N-1 Contingency



Minimum Bus Voltage by N-1 Contingency



Severity Summary



N-1 Results Table

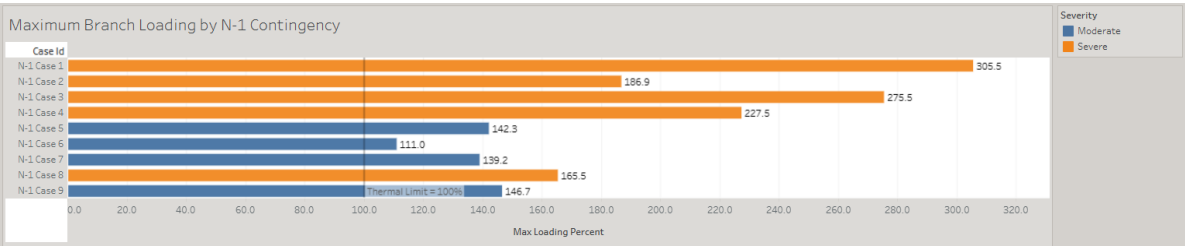
Case Id	Outaged Element	Min Voltage Bus	Max Loading Branch	Severity	Result	Corrective Action	Max Loading...	Min Voltage...
N-1 Case 1	Bruce Generation to GTA...	GTA West	Transformer Station A to ...	Severe	Fail	Reduce Bruce transfer; di...	305.5	0.645
N-1 Case 2	Bruce Generation to Tran...	Western Ontario / Transf...	Transformer Station B to ...	Severe	Fail	Uprate transformer path; ...	186.9	0.851
N-1 Case 3	GTA West to Transformer ...	Toronto Load	Transformer Station A to ...	Severe	Fail	Add or uprate alternate G...	275.5	0.833
N-1 Case 4	Transformer Station B to ...	Toronto Load	Transformer Station A to ...	Severe	Fail	Add or uprate alternate G...	227.5	0.833
N-1 Case 5	GTA Central to Toronto Lo...	Toronto Load	Northern Hydro to Easter...	Moderate	Fail	Add or uprate alternate T...	142.3	0.928
N-1 Case 6	Toronto Load to Eastern O...	Toronto Load	Transformer Station A to ...	Moderate	Fail	Redispatch generation to ...	111.0	0.964
N-1 Case 7	Northern Hydro to Easter...	Ottawa	GTA Central to Toronto Lo...	Moderate	Fail	Increase eastern supply s...	139.2	0.872
N-1 Case 8	Intertie Bus to Eastern On...	Ottawa	GTA Central to Toronto Lo...	Severe	Fail	Increase eastern supply s...	165.5	0.941
N-1 Case 9	Peaker/Battery to GTA Ce...	Ottawa	Transformer Station B to ...	Moderate	Fail	Increase local GTA suppor...	146.7	0.939

Figure 3: Tableau dashboard summarizing N-1 contingency results.

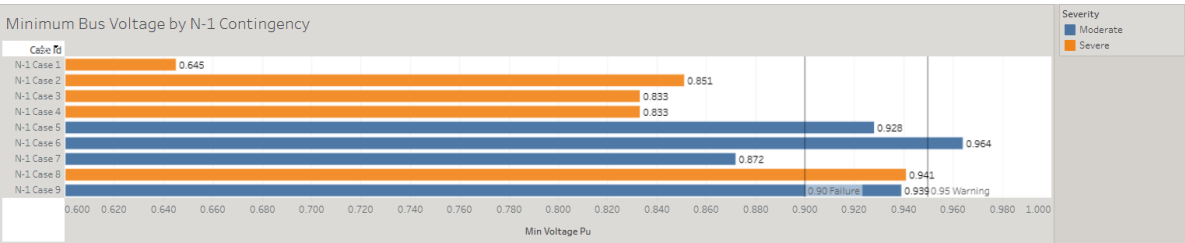
The dashboard shows that 5 of the 9 tested contingencies were severe and 4 were moderate.

The most critical case was N-1 Case 1, which produced both the worst voltage and the highest thermal loading.

The three charts below break the dashboard summary out individually for closer reading.



(a) Maximum branch loading by N-1 contingency case.



(b) Minimum bus voltage by N-1 contingency case.

Figure 4: Individual dashboard chart breakouts.

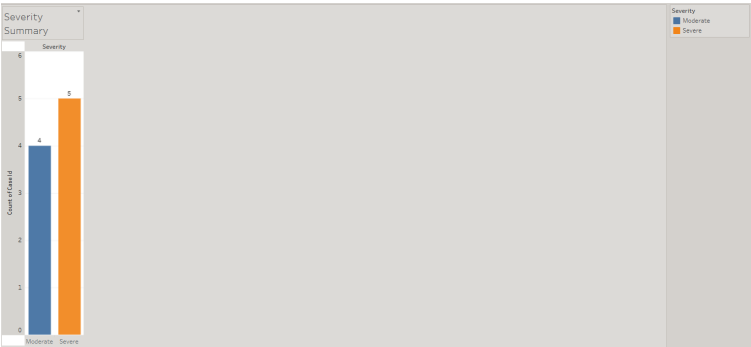


Figure 5: Severity classification summary for N-1 contingency cases.

9 Discussion

The N-1 results show that the system is most vulnerable when major GTA supply paths or transformer corridors are removed from service. Several contingencies caused branch loadings above 100 %, meaning that thermal limits were exceeded. In a real transmission planning environment, these cases would require mitigation through redispatch, switching actions, reactive support, transformer uprating, or additional transmission capacity.

The voltage results also show that some outages caused major voltage depression. Case 1 was especially severe, with a minimum voltage of 0.645 pu. This would not be acceptable in an actual transmission system and indicates that the system lacks sufficient support under that outage condition. The project demonstrates why N-1 analysis is important: a system that may appear acceptable in a base case can still become unreliable after a single critical element is removed.

10 Skills Demonstrated

This project demonstrates the following skills relevant to power system planning and engineering communication:

- Building and validating a multi-bus transmission model in PowerWorld Simulator.
- Solving base case power flow using Newton Raphson methods.
- Performing systematic N-1 contingency testing.
- Identifying and classifying voltage and thermal violations.
- Developing corrective action recommendations.
- Structuring results data in CSV format for downstream analysis.
- Building a Tableau dashboard for engineering communication.

11 Limitations

This project uses a simplified 13-bus model and does not represent the full Ontario transmission network. The values, bus names, and topology were designed for educational and portfolio purposes. The analysis does not include dynamic stability, short circuit analysis, protection coordination, economic dispatch, or full reactive power planning.

Despite these limitations, the project demonstrates the workflow used in transmission planning studies: model creation, power flow solution, contingency testing, violation identification, corrective action planning, and visualization.

12 Future Work

Future iterations of this exercise could be expanded in several directions. A full reactive power and voltage support study, incorporating capacitor banks, SVCs, or generator reactive limits, would strengthen the corrective action recommendations beyond simple redispatch. N-2 or common-mode contingency testing could also be added to evaluate the system's resilience under more severe multi element outages.

The Tableau dashboard could be extended into a live linked workbook that updates automatically as new PowerWorld cases are exported, and the corrective action table could be tied to estimated reinforcement costs to simulate a basic transmission planning cost-benefit comparison.

13 Conclusion

This project completed an N-1 contingency analysis on a simplified 13-bus transmission network using PowerWorld Simulator and Tableau. The study found that 5 of 9 contingencies were severe, with Case 1 producing the worst thermal overload and lowest voltage. The most stressed parts of the system were the GTA supply paths, transformer station corridors, and eastern support paths.

The final deliverables include a PowerWorld model, N-1 case screenshots, structured CSV results, corrective action recommendations, Tableau dashboard visuals, and this technical report. The project demonstrates practical power system analysis, contingency evaluation, engineering judgment, and data visualization.

References

1. NERC Reliability Standards, TPL-001, *Transmission System Planning Performance Requirements*.
2. Bergen, A. R., and Vittal, V., *Power Systems Analysis*, 2nd Edition.
3. PowerWorld Corporation, *PowerWorld Simulator User Guide*.
4. Tableau Software, *Tableau Desktop Documentation*.